

Can we generate portable representations for clinical time series data using LLMs?

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Motivation: The Deployment Loop

The Status Quo: Brittle Clinical ML.

- ⚠️ **Performance Collapse:** Models built at Hospital A degrade at Hospital B due to changing sensor schemas, case mixes, and sampling biases.
- ⚠️ **Three Culprits:** Shift is driven by **Population** (demographics), **Incidence** (prevalence), and **Laboratory** (policy) changes.
- ⚠️ **Inadequate Standards:** OMOP/FHIR solve formatting but lack site-invariant, task-optimized representations for predictive modeling.
- ⚠️ **High Overhead:** Current adaptation requires heavy site-specific data and long recalibration cycles, delaying patient benefit.

The Semantic Thesis.

- 💡 **Portable Inputs:** Mapping records to a site-agnostic semantic space enables models to function with minimal site-specific adaptation.
- 😊 **Clinical Handoffs:** Inspired by narrative summaries that abstract measurement noise and site-specific artifacts while preserving context.
- ⚙️ **Semantic Abstraction:** LLMs allow automatic unit normalization, synonym resolution, and alignment of heterogeneous policies.
- 🔗 **Information Layer:** Using frozen LLMs to bridge irregular numeric ICU time-series and transferable clinical concepts.

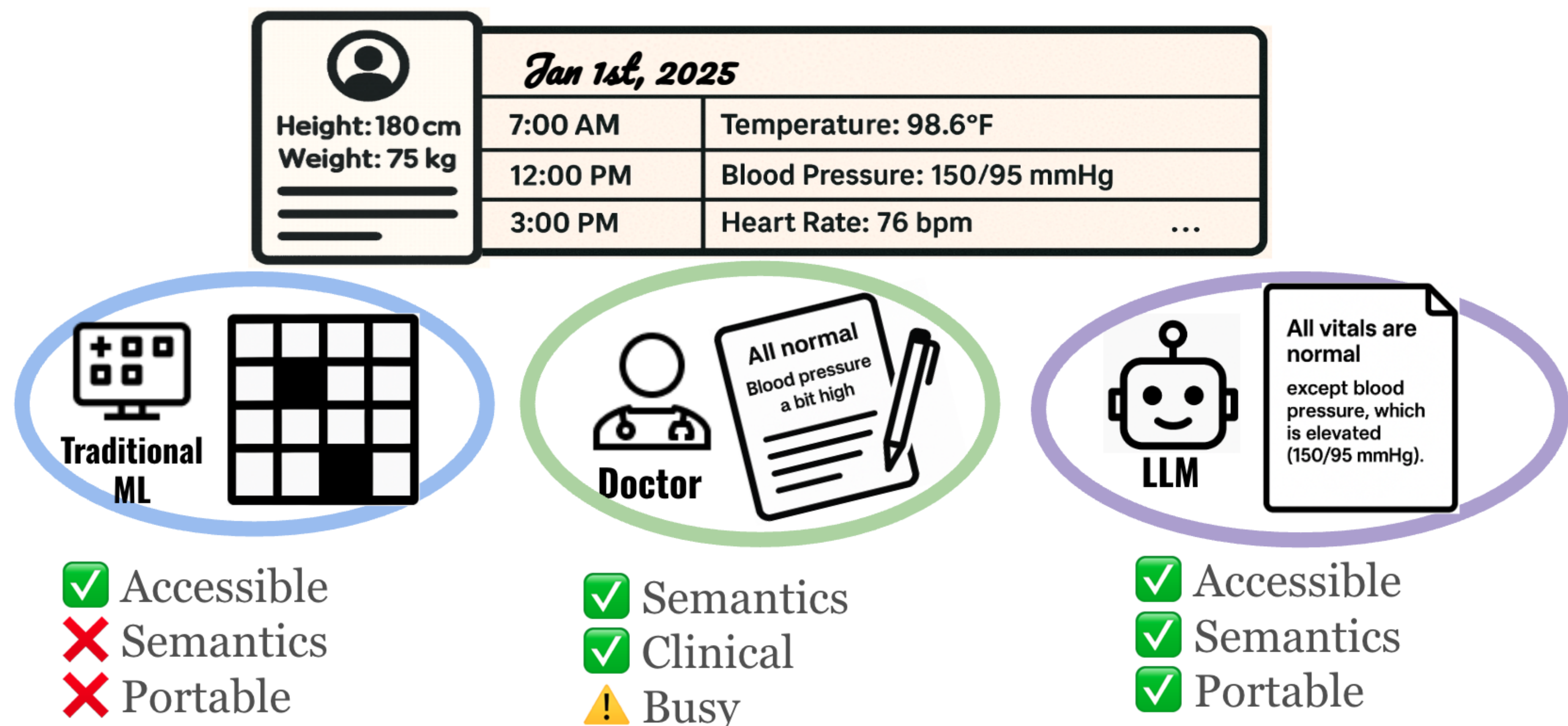


Figure: Motivation for Record2Vec. Numeric imputation loses clinical semantics and limits portability; human handoffs preserve meaning but are costly and variable. LLMs can create handoff-style summaries that retain semantics and provide portable inputs for forecasting and classification.

Our Contributions

- ✅ **Deployment-First Framing:** Rethinking healthcare ML by focusing on site-invariant input representations to reduce per-site engineering.
- ✅ **Record2Vec Pipeline:** A practical summarize-then-embed workflow using frozen LLMs to turn messy ICU data into stable, fixed-length vectors.
- ✅ **Robust Multi-Site Study:** Validated across MIMIC-IV, HiRID, and PPICU cohorts on seven tasks, outperforming purely numeric baselines.
- ✅ **Extreme Data Efficiency:** Adapted models reach target-distribution performance with only **16 labeled samples** via minimal supervision.
- ✅ **Privacy Alignment:** Semantic transformation attenuates demographic attributes while strengthening clinically relevant features for downstream tasks.

The Record2Vec Pipeline

We treat language as an information transformation layer to standardize messy 48h ICU windows.

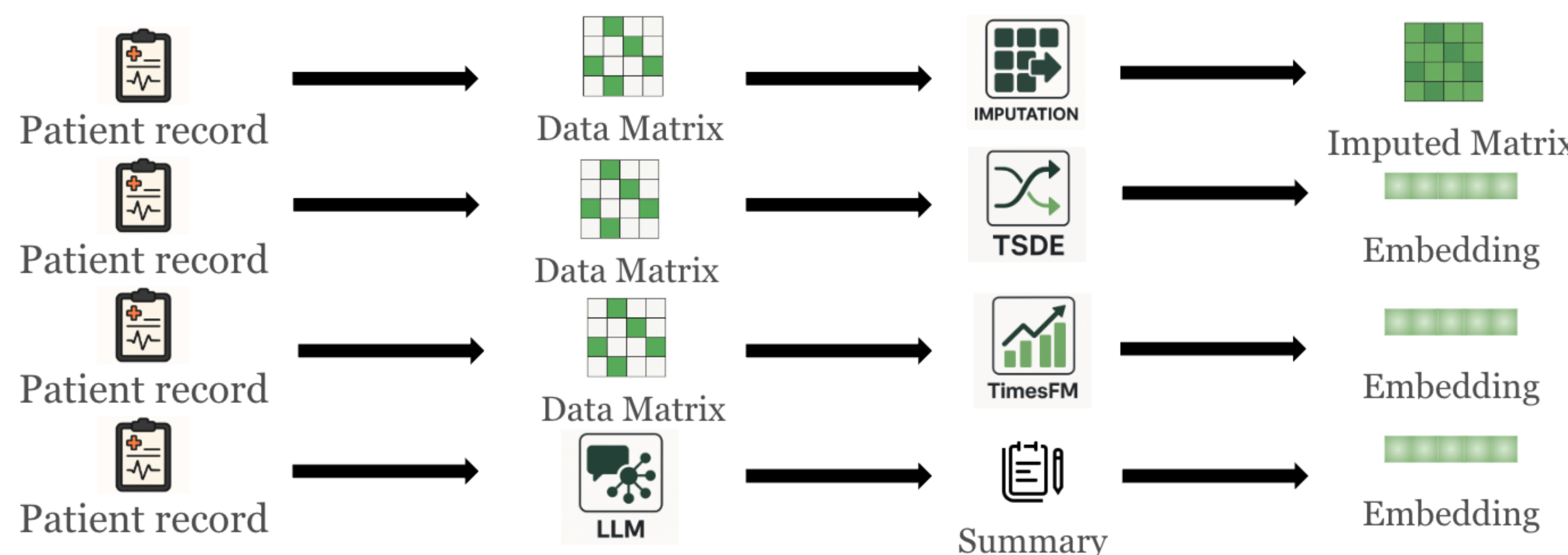


Figure: Methods to generate medical-record representations. Record2Vec uses an LLM summary followed by text embedding to bridge site-specific gaps.

1. Summarization (g_o).

- **Narrative Transformation:** Maps irregular time series to concise clinical summaries, resolving synonyms and normalizing units automatically.
- **Summarizer Regimes:** Tested Gemini-2.0 Flash (General), MedGemma (Medical), and Llama-3.1 (Open-source).

2. Prompt Design & Serialization (RQ4).

- **Serialization:** Raw multivariate records are mapped into a timestamped clinical stream before being fed to the frozen LLM.
- **Structured Prompts:** We evaluate zero-shot, ICD-focused, and trend-focused prompting. ICD-style prompts nudge summaries toward standardized terminology, aiding cross-site transfer.
- **Invariance:** Structured slot-based prompts reduce predictive variance across diverse cohorts without altering mean accuracy.

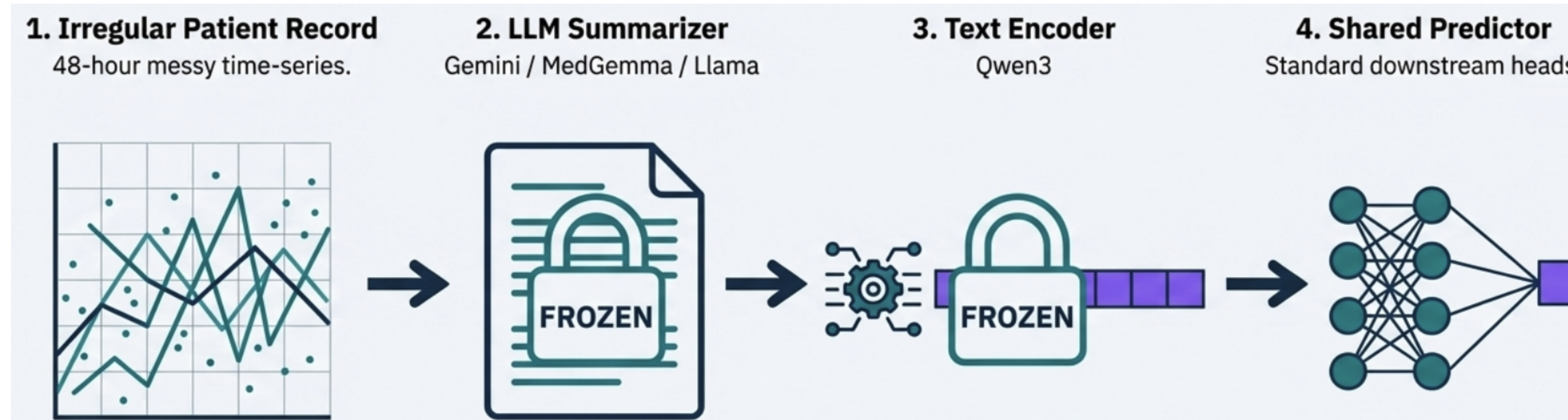


Figure: The Record2Vec Pipeline: A site-agnostic "Summarize-then-Embed" architecture.

3. Embedding (h_o).

- **Fixed-Length Interface:** A frozen text encoder (Qwen3) produces a $z_i \in \mathbb{R}^d$ vector, making downstream models insensitive to missingness patterns.
- **Freezing Strategy:** Both g_o and h_o remain frozen to prevent the representation from overfitting to source-site coding artifacts.

Experimental Setup.

- **Large-Scale Evaluation:** 128,428 total ICU stays across MIMIC-IV (57k), HiRID (32k), and PPICU (39k) with 60–75 merged clinical variables per site.
- **Predictive Tasks:** 7 outcomes including 24h forecasting, mortality, length of stay, drug/lab ordering, and demographic privacy probes.
- **Diverse Baselines:** Compared against grid imputation (Mean/Right-shift), TSDE (Diffusion), TimesFM (Foundation Model), and GenHPF.

Key Results

Unprecedented Portability (RQ2). Record2Vec wins **10 out of 10** cross-site transfer tasks, preventing the performance collapse seen in numeric models.

Data Efficiency (RQ5). Adapted Record2Vec models reach in-distribution performance (trained on 36k samples) with only **16 labeled target samples**.

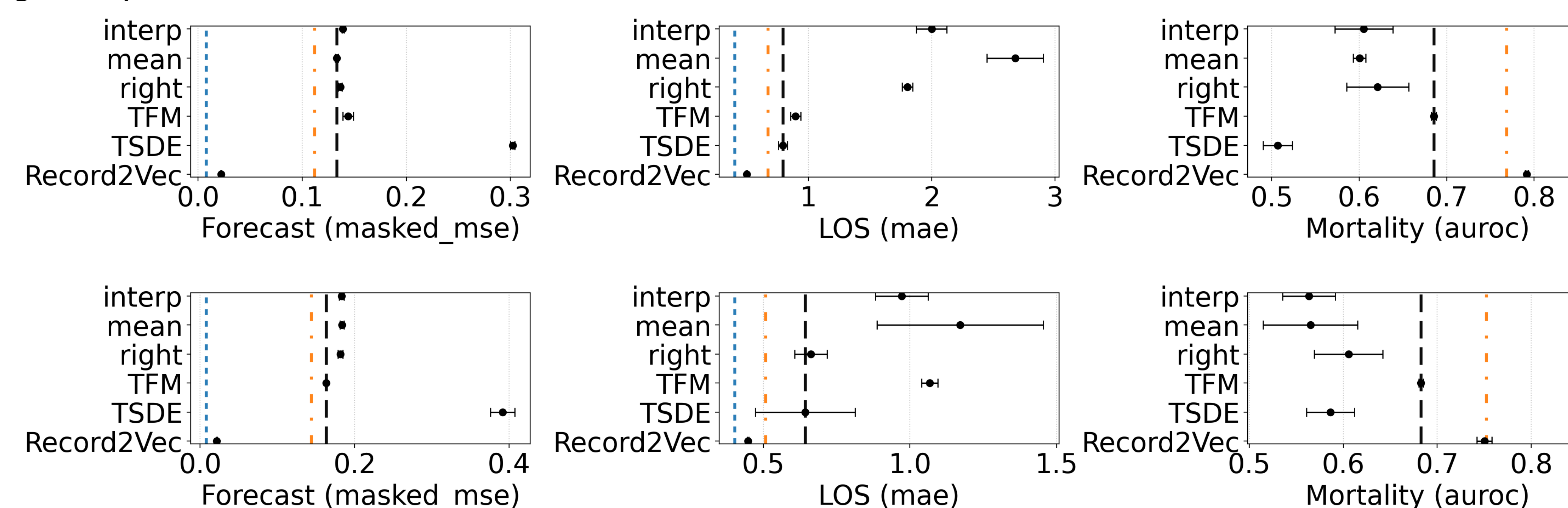


Figure: Few-shot finetuning with 16 labeled target samples for mortality prediction. **blue** = best in-distribution upper bound, **orange** = best pre-finetune result

Privacy & Information (RQ6-7).

- **Demographic Attenuation:** Record2Vec does not increase leakage; gender prediction collapses to chance.
- **Semantic Gain:** Largest improvements in forecasting and drug prediction (+210% in some transfer settings).

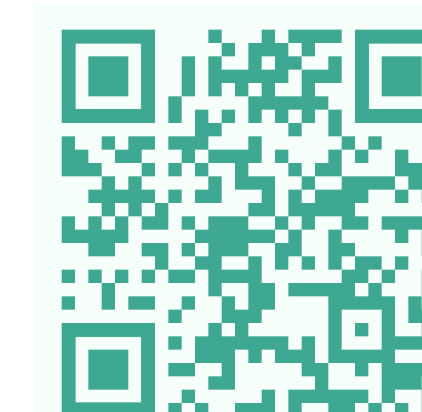
Limitations

- **Resource:** Generating LLM-based representations is more compute-intensive than traditional numeric pipelines, incurring higher latency and token costs.
- **Privacy & Security:** External API processing requires strict IRB/HIPAA controls. Clinical deployment likely requires secure on-prem LLM solutions.
- **Theoretical Gap:** While empirical results are strong, formalizing the conditions for site-invariance using causal tools remains an open research direction.

Conclusion & Future Work

Conclusion. Record2Vec demonstrates that **portable input representations enable portable models**. By using LLMs as a semantic transformation layer, we move clinical ML from site-specific engineering to a shared semantic space that:

- Survives distribution shifts that collapse traditional models.
- Operates effectively with as few as 16 target-site samples.
- Attenuates demographic leakage while preserving clinical utility.



Future Work.

- **Causal Analysis:** Deriving site-invariance conditions via structural causal models.
- **Workflow Integration:** Prospective testing in real-time clinical support systems.
- **Longitudinal Scaling:** Scaling to multi-month records beyond 48h ICU windows.